$$\pi^2 = \frac{9}{2}\prod_{s=1}^{\infty}\left[\frac{\zeta(2s)}{\zeta(2s+2)}\cdot\frac{2^{2s+2}-1}{2^{2s+2}-4}\right]$$

Considering the LHS,

$$\frac{9}{2}\cdot\prod_{s=1}^{\infty}\left[\frac{\zeta(2s)}{\zeta(2s+2)}\cdot\frac{2^{2s+2}-1}{2^{2s+2}-4}\right] = \frac{9}{2}\cdot\prod_{s=1}^{\infty}\frac{\zeta(2s)}{\zeta(2s+2)}\cdot\prod_{s=1}^{\infty}\frac{2^{2s+2}-1}{2^{2s+2}-4} \tag{1}$$

We have, $\displaystyle\prod_{s=1}^{\infty}\frac{\zeta(2s)}{\zeta(2s+2)} = \frac{\zeta(2)}{\zeta(4)}\cdot\frac{\zeta(4)}{\zeta(6)}\cdot\frac{\zeta(6)}{\zeta(8)}\cdots = \zeta(2) = \frac{\pi^2}{6}$ (2)

And, $\displaystyle\prod_{s=1}^{\infty}\frac{2^{2s+2}-1}{2^{2s+2}-4} = \prod_{s=1}^{\infty}\frac{1-2^{-(2s+2)}}{1-2^{-2s}} = \frac{1-2^{-4}}{1-2^{-2}}\cdot\frac{1-2^{-6}}{1-2^{-4}}\cdot\frac{1-2^{-8}}{1-2^{-6}}\cdots$

$$\prod_{s=1}^{\infty}\frac{2^{2s+2}-1}{2^{2s+2}-4} = \frac{1}{1-2^{-2}} = \frac{4}{3} \tag{3}$$

From (1), (2) and (3)

$$\frac{9}{2}\cdot\prod_{s=1}^{\infty}\left[\frac{\zeta(2s)}{\zeta(2s+2)}\cdot\frac{2^{2s+2}-1}{2^{2s+2}-4}\right] = \frac{9}{2}\cdot\frac{\pi^2}{6}\cdot\frac{4}{3}$$

$$\implies \boxed{\frac{9}{2}\cdot\prod_{s=1}^{\infty}\left[\frac{\zeta(2s)}{\zeta(2s+2)}\cdot\frac{2^{2s+2}-1}{2^{2s+2}-4}\right] = \pi^2}$$